

How Resilient Is the HealthCare System in Protecting Patient Data?

By

Sawsan Mounes

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Declaration

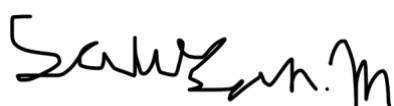
I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

Sawsan Mounes

Date: 21/01/2026

Signed (apply signature below)

A handwritten signature in black ink, appearing to read 'Sawsan Mounes', written in a cursive style.

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I would like to thank my supervisor, Mastaneh Davies, for her guidance, support, and feedback throughout this project. Her advice helped shape the direction of the work and kept me on track at every stage.

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Abstract

The increasing use of digital healthcare systems has improved access to services but has also raised concerns about the security of patient data. This project explores how resilient healthcare systems are in protecting sensitive information and examines public awareness and trust in healthcare applications.

To investigate this, a survey was designed and distributed using Google Forms to collect data on user perceptions of security, privacy, and trust. The questionnaire focused on areas such as data protection, access to medical records, awareness of cyber threats, and the use of security features.

The results show that healthcare applications are widely used, with most participants reporting regular or occasional use. However, many users expressed uncertainty about how their data is stored, who can access it, and how secure it is. Trust was found to be influenced by factors such as transparency, data sharing, and awareness of cyberattacks. In particular, the involvement of third-party services reduced user confidence.

Overall, the findings suggest that while digital healthcare systems are increasingly relied upon, gaps in user understanding and trust still exist. Improving communication, transparency, and user awareness is essential to strengthen confidence in healthcare applications and ensure the protection of patient data.

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1.Introduction

Over the years, healthcare has transitioned from paper-based records to fully digital systems accessed through computers and mobile devices. This digitisation has improved efficiency, accuracy, and accessibility within healthcare services [2], with applications such as the NHS App, Patient Access, mygp, Apple Health, Flo, and Calm now widely used. While these platforms offer clear benefits such as faster access to records and quicker appointments, they also raise significant concerns regarding the security, privacy, and resilience of systems responsible for protecting sensitive patient data. Protecting patient data is essential to maintain patient trust, ensure effective treatment, comply with legal requirements such as HIPAA, and prevent identity theft and fraud.

The healthcare sector has experienced a significant rise in cyberattacks as a result of this increased digital reliance. According to Alder [1], there was a major spike in 2015 when over 112 million healthcare records were exposed, and in 2023 alone, 725 data breaches were reported by the U.S. Department of Health and Human Services (HHS) Office for Civil Rights, exposing over 133 million patient records [1].

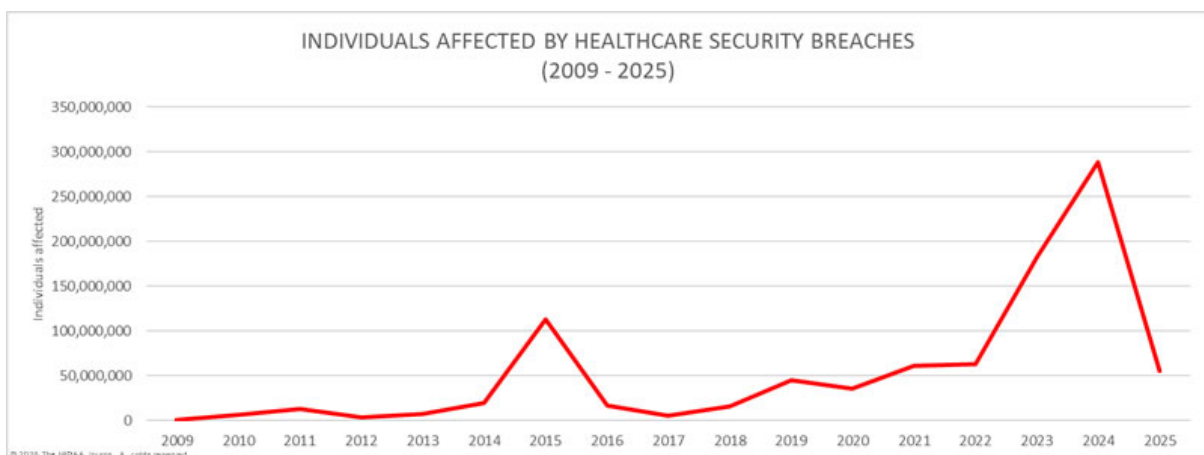


Figure 1: Healthcare Records Exposed by Year [1]

Additionally, hacking-related incidents have become the dominant cause of breaches. Between 2018 and 2023, there was a 239% increase in hacking-related healthcare breaches and a 278% increase in ransomware attacks. In 2019, hacking accounted for 49% of reported breaches, rising to nearly 80% in 2023. These trends highlight the urgent need for stronger cybersecurity resilience within healthcare systems.

The purpose of this report is to examine how resilient healthcare systems are in protecting patient data against modern cyber threats and to assess whether current security measures are sufficient to maintain trust and protect sensitive health information.

Problem Statement and/or Research question

As healthcare organisations move to digital platforms to manage patient records, there are still major risks and threats such as ransomware, data breaches, insider misuse, weak application security, and concerns around patient trust and transparency. This problem affects both patients, users and healthcare employees who access, handle, or share sensitive patient data through healthcare apps and online systems. These risks occur when patient records are stored, accessed, or managed on digital healthcare platforms, especially as more services rely on apps. This research explores how much people trust healthcare apps to keep their data safe, and whether the healthcare system is genuinely well protected against modern cyber threats. It also looks at possible differences in cybersecurity awareness and behaviour across age groups, and highlights that training is not just needed for patients, users, employees, and staff also need consistent cybersecurity training and clear processes to reduce human error and insider risk, so patient data is protected equally.

Aims

- To examine whether the healthcare system is truly secure in protecting patient data from modern cyber threats.
- To explore public trust, confidence, and transparency concerns around healthcare apps and digital patient records.
- To identify how awareness and training (for both patients and healthcare staff) can be improved to reduce risk and protect data equally.

Objectives

- Examine how healthcare systems and apps (e.g., the NHS App) protect patient data.
- Identify best practices for keeping patient information safe.
- Create and analyse a survey to understand public awareness, confidence, and trust in data security.

- Explore how transparency can be improved for patients' medical data (e.g., clearer privacy info and data access controls).
- Suggest ways to make healthcare apps and systems more secure, including training improvements for both users and staff.

2. Background

Healthcare services increasingly rely on digital systems and mobile applications to store and manage patient data. While these technologies improve efficiency and access to care, they also introduce cybersecurity risks due to the sensitivity of healthcare information [9]. High-profile cyber incidents in healthcare have raised concerns about data protection and patient trust [1]. This project is important as it examines public awareness and confidence in the security of healthcare applications, helping to identify whether current systems are trusted and resilient enough to protect patient data.

A widely used framework for evaluating cybersecurity resilience is the CIA triad, which defines three core security properties: Confidentiality, Integrity, and Availability [21]. In the context of healthcare systems, confidentiality refers to ensuring that patient records are accessible only to authorised individuals; integrity means that medical data cannot be altered without detection; and availability ensures that clinical systems remain accessible when needed for patient care. The threats examined in this section including ransomware, phishing, and data breaches each represent direct attacks on one or more of these properties. For example, ransomware primarily targets availability by encrypting systems, while data breaches violate confidentiality, and insider misuse can compromise integrity. This framework, alongside guidance from the National Institute of Standards and Technology (NIST) Cybersecurity Framework, provides a structured lens through which the resilience of healthcare systems can be assessed [22].

Summary of relevant published work

Cyber security threats in healthcare

Due to the increasing reliance on digital platforms for storing patient records and managing sensitive data, there has been a significant rise in cyber threats. Multiple research studies highlight that healthcare organisations are consistently targeted by cyberattacks. Common cyber threats identified in the literature include ransomware attacks, phishing attacks, data breaches, and malware.

Phishing

Phishing is a cyberattack that uses fraudulent emails, messages, phone calls, or fake websites to trick individuals into disclosing sensitive and confidential information. It is a form of social

engineering in which attackers impersonate trusted entities, such as colleagues, managers, authority figures, or well-known organisations. These attacks often encourage victims to take specific actions, including paying fraudulent invoices, opening malicious attachments, or clicking harmful links, which can lead to unauthorised system access or data breaches [4].

Phishing attacks occur frequently and are commonly used as an initial entry point for larger cyber incidents. A real-world example occurred in the United States in December 2018, when Memorial Hospital at Gulfport notified approximately 30,000 patients following a phishing incident that resulted from the compromise of an employee's email account. The breach allowed attackers to gain access to patients' protected health information stored within emails and email attachments.

Memorial Hospital reported that the exposed data included patient names, dates of birth, health insurance information, and details of medical services received at the hospital. A small number of Social Security numbers were also contained within the compromised email account [5].

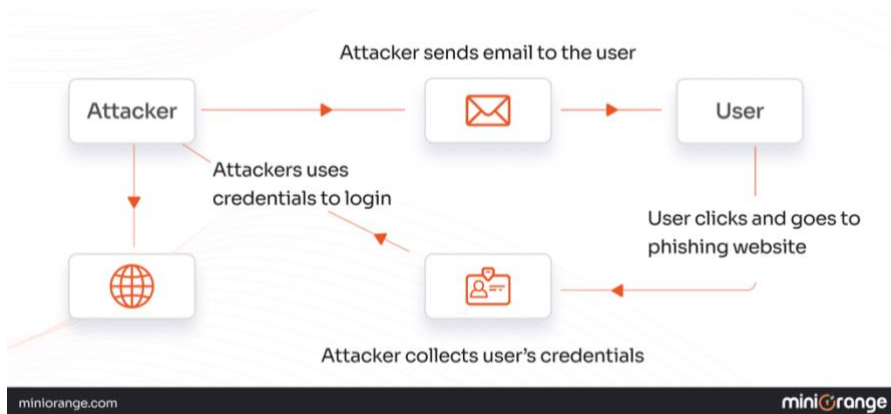


Figure 2: How phishing attacks work [6]

The literature suggests phishing is particularly effective in healthcare because it targets the “human layer” of security, exploiting time pressure and high-volume communication environments. Compared with ransomware (which often relies on technical weaknesses), phishing is frequently linked to limited staff cybersecurity awareness, weak email security controls, and insufficient authentication practices. This helps explain why phishing is often described as an “initial entry point” that can escalate into wider compromise.

Ransomware

Ransomware is a type of malware that prevents users from accessing their devices and the data stored on them, typically by encrypting files. Attackers then demand a ransom payment in exchange for decryption, often instructing victims to make payments using cryptocurrency or to communicate through anonymous email addresses or web pages. In some cases, attackers also threaten to leak stolen data if the ransom is not paid, a tactic known as double extortion [7]. Common attack vectors for ransomware in healthcare include the exploitation of unpatched software vulnerabilities, brute-force attacks against Remote Desktop Protocol (RDP) endpoints, and phishing emails containing malicious attachments. Once an initial foothold is established, attackers typically conduct lateral movement across the network escalating privileges and identifying high-value targets before deploying the ransomware payload [7].

Within healthcare environments, ransomware poses a serious risk to patient data and clinical operations. When systems are encrypted, healthcare organisations may lose access to electronic health records, appointment systems, and diagnostic tools, directly threatening the Availability property of the CIA triad and potentially delaying treatment and compromising patient safety. In addition to service disruption, ransomware attacks can result in the exposure of sensitive patient information, either through data theft or through threats of public disclosure, leading to privacy violations and loss of patient trust.

A well-known real-world example is the wannacry ransomware attack on 12 May 2017 [3], which affected over 200,000 computers across at least 100 countries. The attack encrypted data and demanded ransom payments from victims. Although the NHS was not a direct target, the attack caused significant disruption to healthcare services in the United Kingdom, including cancelled appointments and delayed treatments. Prior to the attack, Microsoft had released a security patch to address the exploited vulnerability; however, many NHS systems had not applied the update, highlighting how outdated systems and poor patch management can increase vulnerability to ransomware attacks.

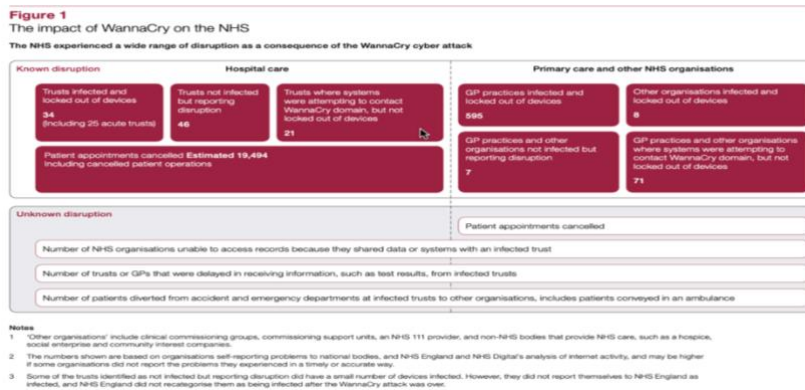


Figure 3: Impact of the wannacry ransomware attack on the NHS [3].

Compared with phishing, ransomware incidents in healthcare are often linked more strongly to technical and organisational weaknesses such as legacy systems, delayed patching, and limited resilience planning. The wannacry example supports the view that even when a fix exists, operational constraints (including low tolerance for downtime) can contribute to delayed updates, which increases exposure to large-scale disruption.

Data breach

A data breach is a security incident in which unauthorised individuals gain access to sensitive or confidential information, including personal data such as Social Security numbers, bank account details, and healthcare records, as well as organisational data like customer records and financial information [8]. In healthcare settings, data breaches can have severe consequences, as they often involve the exposure of medical records containing patients' medical history, diagnoses, insurance details, and identification numbers such as Social Security or NHS numbers. This directly violates the Confidentiality property of the CIA triad. This can also impact healthcare organisations with financial loss. For example, according to an IBM report, the average cost of a data breach in 2019 was \$3.92 million, while a healthcare industry breach typically costs \$6.45 million. In the USA, the average cost of a healthcare data breach (average breach size 25,575 records) has been reported as \$15 million [9].

The long-term impacts of healthcare data breaches may include identity theft, financial fraud, and significant loss of patient trust. Such incidents can also damage the reputation of healthcare organisations and raise serious concerns regarding privacy and data protection within digital healthcare systems.

Healthcare data breaches are commonly enabled by a range of technical weaknesses. The Open Web Application Security Project (OWASP) identifies broken access control, security misconfiguration, and insecure direct object references as among the most critical vulnerabilities in web and application environments [23]. These vulnerabilities are particularly relevant in healthcare, where patient portals and Electronic Health Record (EHR) systems often expose large volumes of sensitive data through web interfaces. Poor data governance such as storing data without adequate encryption at rest or in transit and insufficient monitoring for unauthorised access attempts further increase the risk of breach.

In contrast to ransomware (which often results in operational shutdown) and phishing (which often initiates compromise), data breaches highlight the consequences of weak access control, poor data governance, insecure storage, and inadequate monitoring. The literature consistently links breach severity in healthcare to the sensitivity and volume of centrally stored patient information, meaning that even small security failures can lead to widespread harm.

What weaknesses in healthcare allow cyberattacks to happen in the first place?

Despite increased awareness of cybersecurity risks, healthcare systems continue to display significant vulnerabilities that enable cyberattacks. The literature highlights that these weaknesses are largely linked to outdated infrastructure, poor patch management, weak access controls, and human-related factors.

A major vulnerability is the growing dependence on digital platforms to manage patient information. As noted in the literature, due to the increasing reliance on digital platforms for storing patient records and managing sensitive data, there has been a significant rise in cyber threats. This reliance increases the attack surface of healthcare systems, particularly when security measures are insufficient.

Human factors also play a critical role in system vulnerability. Phishing attacks are frequently described as a common initial entry point for larger cyber incidents, demonstrating how limited cybersecurity awareness and staff training can expose healthcare organisations to attacks. This was evident in the Memorial Hospital at Gulfport incident [5], where attackers accessed patients' protected health information stored within emails and email attachments.

Outdated systems and poor patch management further increase vulnerability. During the wannacry ransomware attack, many NHS systems had not applied the update, despite a security patch being

available, allowing the attack to spread rapidly. Additionally, healthcare organisations' low tolerance for downtime can delay essential updates, meaning systems remain exposed. As a result, attacks may lead to loss of access to electronic health records, appointment systems, and diagnostic tools, directly impacting patient care.

Finally, when large amounts of patient data are stored in central systems, security failures can have serious consequences. Data breaches often expose medical records containing personal details such as medical history, diagnoses, insurance information, and identification numbers. This can result in identity theft, financial harm, and a significant loss of trust between patients and healthcare organisations.

These threats and weaknesses justify the need for stronger protection and clearer reassurance around how patient data is secured on digital platforms. In particular, the literature indicates that patient trust can be undermined when breaches occur and when individuals do not understand how their information is protected. Therefore, this project uses these findings to inform the design requirements of the proposed solution by focusing on security and privacy controls that reduce the likelihood of unauthorised access and improve transparency for users.

These issues also raise legal, ethical, and professional concerns because healthcare data is highly sensitive. Data breaches and unauthorised access can result in privacy violations and non-compliance with data protection requirements (e.g., confidentiality obligations and data protection principles). As a result, the literature supports the need to consider privacy-by-design, secure access control, and responsible handling of patient information throughout the project's design and

Security weaknesses in health care apps

While healthcare applications provide fast and convenient access to services, they are not always designed with strong cybersecurity protections. The literature suggests that many mobile health applications pose security and privacy risks when sensitive data is collected, stored, or shared without adequate safeguards [10]. Additionally, healthcare applications such as Apple Health, Amazon Medical, and Google Cloud Healthcare rely heavily on application programming interfaces (apis) to enable communication and data exchange between systems. These apis are used to transfer medical records, update patient information, and share sensitive data across platforms, which can introduce security risks if not properly protected

A key API vulnerability in healthcare is Broken Object Level Authorisation (BOLA), ranked number one in the OWASP API Security Top 10 [23]. BOLA occurs when an API fails to verify whether a user is authorised to access a specific data object, potentially allowing one patient to retrieve another's health record by simply modifying a request. Additional risks include insecure data transmission without adequate encryption, and excessive data exposure where APIs return more information than necessary both of which conflict with GDPR data minimisation requirements [18].

As a result of this increased reliance on APIs, research indicates that third-party providers are frequently targeted because they are often granted excessive access privileges rather than following the principle of least privilege. As a result, cyberattacks commonly involve the compromise of privileged accounts, making effective privileged access management (PAM) critical. These attacks often occur through compromised remote access tools, such as VPNs, due to phishing or the absence of multi-factor authentication.

Security weaknesses in healthcare applications are not only software-related but also user-related, as unauthorised access frequently results from weak authentication practices, poor access control, and limited user security awareness. Implementing stronger security measures, such as access restrictions and two-factor authentication, is essential to reduce the risk of unauthorised access to healthcare platforms [11].

The NHS App provides built-in identity verification through NHS Login, which uses the Fast Identity Online 2 (FIDO2) authentication standard [12]. This standard removes the need for traditional passwords, reducing the risk of credential theft. In contrast, Patient Access requires users to install external authenticator applications to enable additional security layers during sign-in [13]. The NHS App therefore demonstrates a stronger security model, as reliance on third-party authenticator applications may introduce additional risks, since previous research indicates that third-party services are often granted broader system access than necessary [10].

Patient trust and digital health

Patient trust plays a key role in the use of digital healthcare systems. Research shows that trust is especially important in healthcare settings, as patients rely on organisations to manage sensitive personal data. As stated in the literature, “trust has long been recognized as an important factor... particularly important in the context of healthcare” [16].

Trust also affects whether users are willing to share their personal health data. Previous research highlights that “distrust may reduce patients’ willingness to accept others’ access to and use of their personal data” [16]. This shows that concerns around privacy and data security can directly influence how comfortable users feel when using healthcare applications.

However, the study also found that “a large majority (81.9%) of respondents revealed high levels of trust in the ability of healthcare to protect electronic patient data” [16]. This suggests that trust can be high in certain contexts, particularly when systems are well established. A strength of this study is that it provides clear statistical evidence, making the findings more reliable. However, one limitation is that the study is based on a specific population, meaning the results may not fully represent all users.

In addition to this, wider industry research highlights the serious impact of data breaches on organisations and trust. The global average cost of a data breach reached USD 4.44 million, with healthcare being the most expensive industry affected [17] . This shows how valuable and sensitive healthcare data is, and why it is a major target for cyberattacks.

Furthermore, trust also influences how users perceive risk. The research shows that “those with low levels of trust also perceived risks of unauthorized access... to be higher” [16]. This is important for this project, as it helps explain why some users feel unsure or concerned when using healthcare applications.

Overall, the literature shows that trust is strongly linked to privacy, transparency, and data security. Both academic research and industry reports highlight that without strong protection and clear communication, users may feel uncertain about using digital healthcare systems. These findings are directly relevant to this project, as they support the need to investigate how users understand and trust healthcare applications.

Legal and regulatory framework

Healthcare data is highly sensitive, meaning it must be protected by strict legal and regulatory frameworks. These frameworks ensure that personal data is handled securely, ethically, and transparently, which is essential for maintaining patient trust in digital healthcare systems.

One of the main regulations is the General Data Protection Regulation (GDPR), which applies in the UK and Europe. GDPR requires organisations to protect personal data, ensure it is only used for specific purposes, and give individuals control over their information, including the right to access or delete their data [18]. In the UK, this is supported by the Data Protection Act 2018, which provides additional legal requirements for how personal data is processed and stored [19].

In addition, the Health Insurance Portability and Accountability Act (HIPAA) is a key regulation used in the United States to protect patient information. HIPAA requires healthcare organisations to ensure confidentiality, integrity, and security of patient data, particularly when it is stored or shared digitally [20].

These regulations represent examples from two different countries, showing how data protection in healthcare is addressed on an international level. Although the specific laws may differ, they share similar goals of protecting patient data, ensuring privacy, and maintaining trust in digital healthcare systems.

These regulations also emphasise the importance of informed consent, meaning users must be clearly informed about how their data will be used before agreeing to share it. This links closely to privacy by design, where systems are built to protect user data from the beginning.

These legal frameworks are directly relevant to this project, as they address many of the issues identified in the findings, including trust, data sharing, and user awareness. If healthcare applications fail to follow these regulations, it can lead to data breaches, legal penalties, and loss of user trust.

Human factors

Human behaviour is a significant contributor to cybersecurity vulnerabilities in healthcare settings. The cybok Human Factors Knowledge Area highlights that non-compliance with security practices is often caused by people facing a choice between doing what is right by security and maintaining their productivity, with most choosing productivity [24]. In healthcare environments, this is particularly relevant as staff working under time pressure may bypass security controls such as skipping authentication steps or sharing login credentials in order to prioritise patient care. This pattern is referred to as security fatigue, where the accumulation of security-related tasks over time increases the likelihood that they will be bypassed [24]. In other words, the more security steps staff are asked to complete, the more likely they are to start ignoring them, which creates openings for attackers.

The way users interact with healthcare applications also affects security and privacy outcomes. Users who do not understand how their data is handled, or who find security features difficult to use, are less likely to engage with them effectively. The NCSC emphasises that multi-factor authentication (MFA) provides significantly stronger protection than passwords alone, and recommends organisations implement techniques that give better protection against phishing attacks [25]. MFA works by requiring users to verify their identity through a second method such as a code sent by SMS, a biometric scan, or an authenticator app meaning that even if a password is stolen, an attacker cannot access the account without the second factor. Without adequate awareness and training, however, users may not enable or correctly use these features, leaving their accounts vulnerable.

The NCSC's Cyber Essentials Plus scheme sets out a standard of cybersecurity controls that organisations, including healthcare providers, are expected to implement to protect against common cyber threats [25]. It requires controls such as MFA, access restrictions, and secure configuration to be in place and independently verified meaning an external body checks that the controls actually work, rather than an organisation simply claiming they do. The failure to implement such controls increases the risk of human-related breaches, as stronger authentication methods such as FIDO2 provide phishing resistance and theft resistance [25], yet many healthcare systems continue to rely on weaker alternatives.

Overall, the literature demonstrates that technical security measures alone are insufficient to protect healthcare systems. Human factors including staff awareness, training, the usability of security tools, and organisational culture play a critical role in determining whether security controls are effective in practice. Addressing these human elements is therefore essential to improving the resilience of healthcare systems and ensuring patient data is consistently protected.

Competitor Analysis/ Related work

The related work demonstrates that while existing studies and reports focus on cybersecurity risks, breach impacts, and public attitudes toward health data, there is limited research examining public awareness and trust in everyday healthcare applications. Several of these studies adopt survey-based approaches involving members of the public, which aligns with the artefact used in this project. This project builds on that approach by gathering public responses on healthcare app security, transparency, and confidence

Table 1: Related work and competitor analysis

Source	Type	Focus	Method use	About
IBM – Cost of a Data Breach Report	Industry report	Impact of data breaches in healthcare	Analysis of global breach data	Highlights financial and trust impacts of healthcare data breaches, supporting the importance of cybersecurity resilience examined in this project
NHS Digital – Public Attitudes to Data in the NHS	Government survey report	Public trust and attitudes toward NHS data use	National public survey	Provides insight into public trust and transparency concerns, informing survey questions on confidence and data protection
Belfrage et al. (2022)	Academic research paper	Trust and digital privacy in healthcare	Cross-sectional questionnaire survey	Demonstrates how survey-based methods can measure public trust, <u>similar to the methodology used in this project</u>
The Health Foundation – Public Views on Health Technologies	Policy analysis report	Public perceptions of health technologies and data	Public opinion analysis	Highlights concerns about data sharing and transparency, supporting the relevance of this project's focus on user awareness
PMC Article – Security & Privacy in Mobile Health Apps	Academic review	Security and privacy risks in mobile health apps	Literature review	Identifies common app-level security weaknesses, providing technical context for survey topics
HHS – Healthcare Industry Cybersecurity Practices (HICP)	Government guidance	Cybersecurity risks and controls in healthcare	Risk assessment & best practice framework	Supports discussion of threats such as phishing and ransomware, reinforcing the need to assess public confidence in healthcare security

3.Tools and Methods

Rationale for the technical tools selected

The primary technical tools used in this project are online survey platforms and data analysis software. Google Forms was selected to design and distribute the questionnaire due to its accessibility, ease of use, and ability to securely collect and export data. The platform supports efficient distribution to a wide audience and allows anonymous participation, which is essential given the sensitive nature of healthcare related topics. In addition, Google Forms is widely used in everyday contexts, making it familiar and easy to trust for many participants.

Spreadsheet software, such as Microsoft Excel, is used to organise and analyse survey responses. This tool enables quantitative data to be summarised, compared, and visualised effectively, supporting the identification of trends across different groups. In addition to Excel, Python was considered as an alternative option for creating data tables and performing data analysis.

Alternative tools such as Qualtrics and surveymonkey were considered; however, these platforms were not selected due to restricted access and additional features that were not required for the scope of this project. The selected tools were therefore considered the most appropriate in terms of usability, cost, and alignment with the project aims.

Rationale for the research methods used

This project adopts a quantitative, survey-based research methodology to examine public awareness, confidence, and trust in the security of healthcare applications. Survey research is widely used in healthcare and cybersecurity studies as it enables the collection of structured and measurable data from many participants.

Quantitative surveys are particularly suitable for identifying patterns and differences in user perceptions, such as variations across age groups or levels of digital experience. The use of closed ended and Likert-scale questions allows responses to be analysed objectively and compared effectively

This methodology is appropriate for the project as it supports the research aims while avoiding the collection of sensitive health data, making it both ethically suitable and effective for generating evidence-based insights into patient trust and healthcare app security.

Project management and Product lifecycle

Project management for this project was carried out using simple planning tools, including a personal notes application and a calendar. Notes were used to record ideas, literature findings, supervisor feedback, and weekly tasks, while a calendar was used to plan deadlines and schedule different stages of the project, such as the literature review, survey design, and data analysis. This approach supported an iterative project lifecycle, allowing sections of the work to be reviewed and changed over time. Using basic planning tools was suitable for this research-based project, as it helped maintain organisation and progress without adding unnecessary disruptions.

Legal, Ethical, and Professional issues

This project involves collecting survey responses from the public about healthcare app security, which raises important legal, ethical, and professional considerations.

From a legal perspective, the project follows data protection principles such as GDPR and HIPAA. Although no medical or personal health data is collected, all survey responses are handled anonymously. Participants are informed about how their data will be used, and no identifying information is requested.

Ethically, informed consent is obtained before participants complete the survey. The purpose of the research is clearly explained, participation is voluntary, and respondents can choose not to answer any question. The survey avoids collecting sensitive health details to reduce risk. Ethical approval was discussed with the supervisor, and the ethics form is included in the appendix.

Professionally, secure data handling was considered throughout the project. It was discussed that deleted data can sometimes still be recovered using digital forensic techniques. To reduce this risk, survey data is stored securely, and any removable storage used would be fully reformatted after use. These steps ensure responsible handling of data and reflect good cybersecurity practice.

Risk register

Several potential risks were identified during this project, mainly relating to data collection, participant engagement, and ethical considerations. One key risk was a low survey response rate, which could limit the reliability of the findings; this was mitigated by keeping the questionnaire short, clear, and easy to complete, and by distributing it online to reach a wider audience. There was also a risk that participants might misunderstand cybersecurity-related questions, so simple language was used, and questions were reviewed before distribution.

Data privacy was another important risk, as survey responses could be accessed or mishandled if not stored securely. This risk was reduced by avoiding the collection of personal or health data, ensuring responses remained anonymous, and storing data securely. It was also considered that digital data may remain recoverable even after deletion due to residual data stored in memory. To reduce this risk, any removable storage used for the project would be fully reformatted after use to ensure that data could not be recovered. Ethical risks, such as participants not fully understanding the purpose of the study, were addressed by including a clear consent statement and making participation voluntary.

4.Requirements/Specifications

Personas

Persona 1: Young Healthcare App User

Name: user 1

Age: 22

Background: University student who regularly uses the NHS App to book GP appointments and check prescriptions.

Goals:

- Quick access to healthcare services
- Secure login such as Face ID

Concerns:

- Unsure how secure healthcare apps are
- Concerned about healthcare data breaches

Persona 2: Older Healthcare User

Name: user 2

Age: 55

Background: Occasionally uses healthcare apps to check appointments or prescriptions but does not fully understand how digital health systems manage patient data.

Goals:

- Simple and trustworthy healthcare platforms
- Clear explanations of how data is stored

Concerns:

- Fear of identity theft or misuse of medical data
- Lack of transparency about who can access patient records

These personas were used to guide the design of the survey, particularly focusing on trust, transparency, and awareness of cybersecurity risks.

User Stories

- As a healthcare app user, I want to understand how my medical data is protected so that I can trust healthcare apps.
- As a survey participant, I want the questionnaire to be short and anonymous so that I feel comfortable sharing my opinions.
- As a researcher, I want to collect responses from different age groups so that I can compare awareness and trust levels.

User Journey

The process of completing the survey is as follows:

1. The participant receives the survey link
2. The participant reads the participant information and consent section
3. The participant confirms they are over 18 and agrees to take part
4. The participant answers background questions such as age group and app usage
5. The participant answers questions about awareness of healthcare data security
6. The participant answers questions about trust in healthcare applications
7. The participant submits the survey

5. Solution design

This section explains the design of the technical artefact used in this project. The artefact developed for this research is an online survey created using Google Forms. The purpose of the survey is to explore public awareness, trust, and views on cybersecurity in healthcare applications.

The survey was designed to be clear, short, and easy to complete. It takes around 5 to 10 minutes and includes approximately 15 to 17 questions.

The questionnaire is divided into several sections. The first section includes the participant information and consent form, where participants confirm they are over 18 and agree to take part in the study.

The second section collects basic demographic information, such as the participant's age group and whether they use healthcare apps.

The following sections focus on awareness and trust in healthcare app security. These questions look at whether participants understand how healthcare data is stored, whether they are aware of cyber threats such as data breaches or ransomware, and how much they trust healthcare applications to protect their personal data.

Most of the questions use a Likert scale, ranging from strongly disagree to strongly agree, to measure participants' opinions. This makes the responses easier to analyse and compare.

The final question asks participants what changes would increase their trust in healthcare applications, such as stronger login security, better transparency, or more control over data sharing. Google Forms was chosen because it allows anonymous participation, is easy to share, and automatically collects responses for later analysis.

6. Solution implementation

This section explains how the artefact for this project was created and implemented. The artefact used in this research is an online survey designed to explore public awareness, trust, and views on cybersecurity in healthcare applications.

The survey was created using Google Forms, which made it easy to design, share, and collect responses while keeping participants anonymous.

Creating the Survey

The first step in the implementation process was creating the questionnaire using Google Forms. This platform was chosen because it is easy to use, accessible on different devices, and allows anonymous data collection.

The survey followed the structure outlined in the previous section. Different types of questions were included, such as multiple choice, checkbox, and Likert scale questions. These made it easier for participants to respond and helped produce structured data that could be analysed later.

The survey was divided into several sections, including participant consent, demographic questions, awareness of healthcare app security, trust in healthcare applications, and security behaviours.

Implementing the Consent Process

One of the main challenges during the implementation stage was the consent form. At first, it was created as a separate document that participants had to read and sign before doing the survey (see appendix A). However, this could have required people to share personal information and might have put some participants off.

To fix this, the consent process was added directly into the Google Forms survey (see appendix B). Participants were asked to read the information and confirm that they were 18 or older, understood the study, and agreed to take part before continuing.

This made the process much simpler and kept the survey anonymous, while still meeting ethical requirements.

Developing the Survey Questions

The survey questions were based on the aims of the project. They were designed to explore different areas related to healthcare cybersecurity, including:

- whether participants use healthcare applications
- awareness of how healthcare data is stored and shared
- understanding of cybersecurity threats such as phishing and ransomware
- trust in healthcare applications to protect personal health data

Most of the questions used a Likert scale, allowing participants to show how much they agree or disagree with each statement. This made it easier to understand people's views and analyse the results clearly.

Testing the Survey

Before sharing the survey, it was tested to make sure all the questions worked properly, and that the layout was clear. The wording was also checked to ensure it was easy to understand, so participants could complete it without confusion.

The survey was designed to take around 5 to 10 minutes to complete, keeping in mind that many people have a short attention span, which helped make it more likely that they would finish it.

Survey Distribution and Data Collection

Once the survey was ready and tested, it was shared using the link from Google Forms. Participants could access it on any device with internet access.

All responses were automatically collected and stored in Google Forms. The platform also showed simple charts and summaries, which made it easier to spot patterns and trends. The data could also be downloaded into a spreadsheet for further analysis.

Ethical Considerations During Implementation

Throughout the implementation process, ethical considerations were prioritised. The survey was designed to ensure that no personal or medical information was collected, allowing participants to remain anonymous.

Participants were clearly informed about the purpose of the research, how their responses would be used, and that participation was voluntary. Participants also had the option to exit the survey at any time if they chose not to continue.

All collected data will only be used for academic research purposes and will be securely deleted once the project and assessment process have been completed.

7. Solution testing/ evaluation

Participant Overview

A total of 61 participants completed the survey. All participants confirmed they were over the age of 18 and gave informed consent before taking part.

The majority of respondents were aged 18 to 24 (78.3%), with smaller numbers from other age groups. This shows that the sample is mostly made up of younger users, particularly students and young adults.

What is your age group?

60 responses

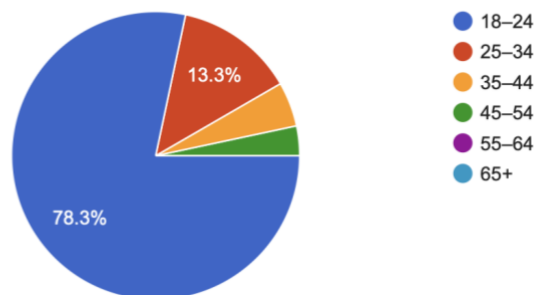


Figure 4: Age group distribution of participants

As shown in Figure 4, this indicates that the sample is more towards younger users, particularly students and young adults, which may limit how well the findings apply to older populations.

Use of Healthcare Applications

Do you use healthcare-related apps (e.g. NHS App, Patient Access, pharmacy or health apps)?

60 responses

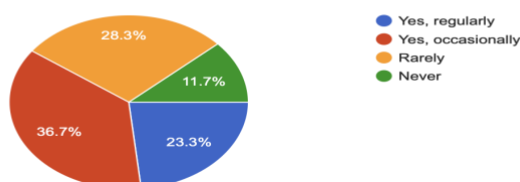


Figure 5: Use of healthcare applications

The results show that 36.7% of participants use apps occasionally, 23.3% use them regularly, 28.3% rarely use them, and 11.7% never use them.

As shown in Figure 5, healthcare applications are widely used among participants, with most people using them either regularly or occasionally. This shows how digital healthcare services are becoming part of everyday life and highlights the importance of having strong cybersecurity in place to protect user data.

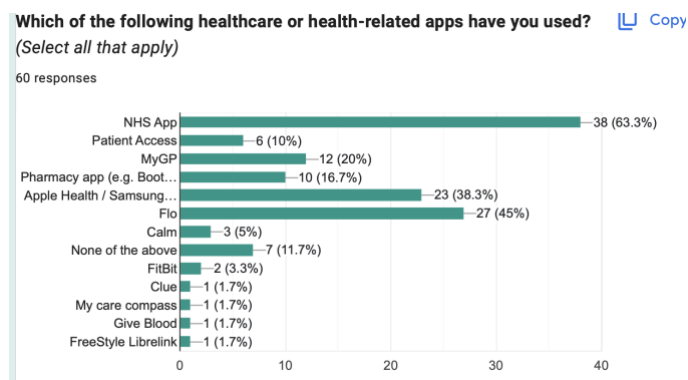


Figure 6: Most commonly used healthcare apps

The most commonly used app was the NHS App (63.3%), followed by Flo (45%) and Apple Health / Samsung Health (38.3%).

This shows that both public healthcare platforms and commercial health apps are widely used by participants. The high use of the NHS App suggests that many people rely on national healthcare services, while the popularity of fitness and health tracking apps shows how digital tools are becoming part of everyday health management.

These findings highlight the importance of strong cybersecurity across both public and private healthcare platforms, as many users depend on them to store and manage sensitive health information.

Awareness of Healthcare Data and Security

I understand what information is stored in my digital health record.

60 responses

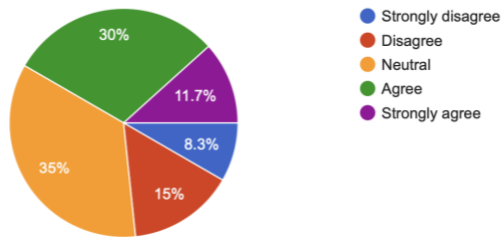


Figure 7: Understanding of stored health data

Only 41.7% of participants agreed or strongly agreed that they understand what information is stored in their digital health record, while 35% remained neutral. A further 23.3% disagreed or strongly disagreed, showing that many participants do not fully understand this.

This suggests that a lot of users are not clear on how their health data is stored or managed. The high number of neutral responses also shows that many people are unsure, which could be due to limited transparency or lack of awareness about how data is handled.

This lack of understanding could affect user trust, as people may feel less confident using healthcare applications if they are unsure how their sensitive information is being stored and protected.

I am aware that healthcare apps may share data between different healthcare systems or services.

59 responses

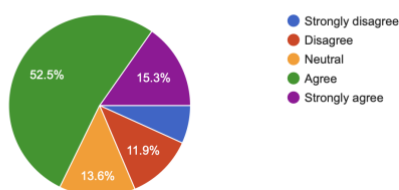


Figure 8: Awareness of data sharing

As shown in Figure 8, 67.8% of participants agreed or strongly agreed that healthcare applications share data between different systems or services. However, 13.6% remained neutral, and a smaller proportion disagreed, showing that not all users are fully aware of how their data is shared.

This suggests that while most participants have a general awareness of data sharing in healthcare systems, some users are still unsure. The neutral and negative responses show that not everyone fully understands how their data moves between different platforms.

This highlights the need for clearer communication and better transparency around how data is shared, as this could help improve user trust and confidence in healthcare applications.

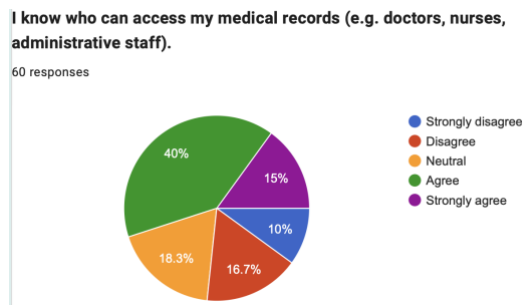


Figure 9: Knowledge of who can access records

As shown in Figure 9, 55% of participants agreed or strongly agreed that they know who can access their medical records, while 18.3% remained neutral. However, 26.7% of participants disagreed or strongly disagreed, showing that a good number of users are still unsure about who can access their information.

This suggests that even though over half of participants have some idea about who can access their records, many people still lack clear understanding. The mix of neutral and negative responses shows uncertainty and gaps in knowledge around access to personal data.

This lack of awareness could raise concerns about transparency and may affect user trust, as people may feel less comfortable using healthcare apps if they are unsure who can see their sensitive information.

Awareness of Cybersecurity Threats

I understand what a healthcare data breach is.

60 responses

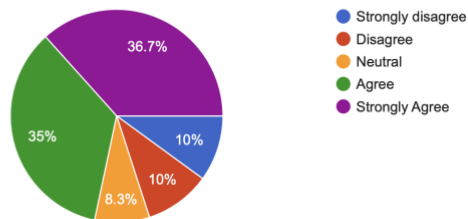


Figure 10: Awareness of cyber threats

As shown in Figure 10, 71.7% of participants agreed or strongly agreed that they understand what a healthcare data breach is. However, 8.3% remained neutral, while 20% of participants disagreed or strongly disagreed, showing that not everyone fully understands this concept.

This suggests that although most participants have some understanding of data breaches, there is still a clear gap in knowledge for some users. The mix of neutral and negative responses shows that some people are unsure or not fully confident in their understanding.

This lack of understanding could make it harder for users to recognise potential risks and may affect how aware they are of cybersecurity issues in healthcare systems.

I am aware that cyberattacks such as phishing or ransomware can affect healthcare services.

50 responses

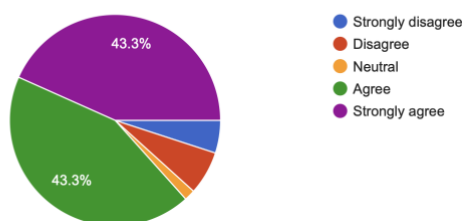


Figure 11: Understanding of data breaches

As shown in Figure 11, 86.6% of participants agreed or strongly agreed that cyberattacks such as phishing and ransomware can affect healthcare services. Only a small number of participants remained neutral or disagreed.

This shows that most people are aware of cybersecurity threats in healthcare and understand that these attacks can have serious impacts. The high level of agreement suggests that users recognise the risks linked to cyberattacks.

This awareness may be influenced by media coverage and general discussions around cybersecurity. However, even though people are aware of these threats, earlier results show that many do not fully understand specific issues like data breaches or who can access their data.

Trust in Healthcare Applications

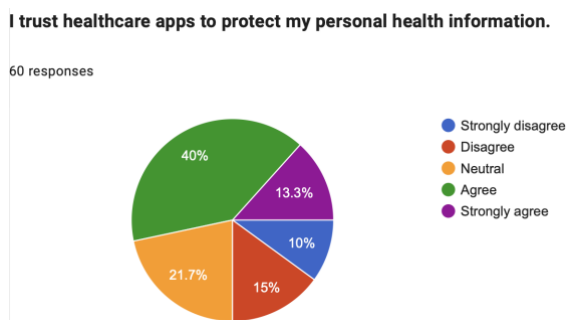


Figure 12: Trust in healthcare apps

As shown in Figure 12, 53.3% of participants agreed or strongly agreed that they trust healthcare applications to protect their personal health information. However, 21.7% remained neutral, while 25% of participants said they do not trust them.

This shows that even though over half of participants do trust healthcare apps, a lot of people are still unsure or don't fully feel confident about how their data is being protected. The mix of neutral and negative responses suggests that trust is not fully there yet.

Overall, this suggests that even though healthcare apps are widely used, people still have concerns about data security and privacy, which affects how much they trust them.

Hearing about healthcare cyberattacks has reduced my trust in healthcare apps.

59 responses

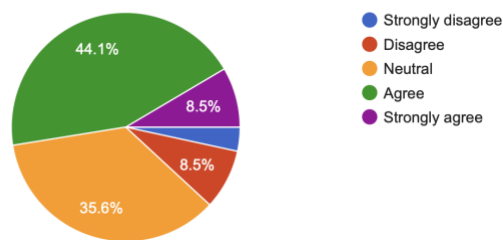


Figure 13: Impact of cyberattacks on trust

As shown in Figure 13, 52.6% of participants agreed or strongly agreed that hearing about healthcare cyberattacks has reduced their trust in healthcare applications. In comparison, 35.6% of participants remained neutral, while only a small proportion disagreed.

This shows that cyberattacks clearly affect how people feel about using healthcare apps, with over half saying their trust has gone down. The high number of neutral responses also suggests that some people are unsure, or may not fully understand how these attacks actually affect them.

Overall, this highlights how important it is for healthcare apps to not only be secure, but also to clearly show users how their data is being protected so they feel more confident using them.

I trust NHS healthcare apps more than private healthcare or health and fitness apps.

60 responses

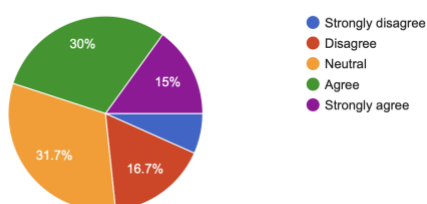


Figure 14: Trust in NHS vs private apps

As shown in Figure 14, 45% of participants agreed or strongly agreed that they trust NHS healthcare applications more than private healthcare or health and fitness apps. However, 31.7% remained neutral, while approximately 23% of participants disagreed or strongly disagreed.

This suggests that although a fair number of participants trust NHS applications more, this trust is not particularly strong. The high percentage of neutral responses indicates uncertainty, suggesting that many users do not have a clear preference between NHS and private platforms.

These findings show that while people tend to trust NHS services slightly more, that trust is not fully established. Both NHS and private healthcare applications may need to improve transparency and clearly communicate how they protect user data to increase confidence.

I trust healthcare apps less when they rely on third-party services to manage authentication or data processing.

60 responses

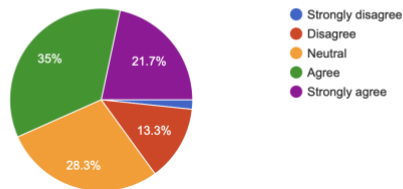


Figure 15: Trust in healthcare apps when third-party services are used

As shown in Figure 15, 56.7% of participants agreed or strongly agreed that they trust healthcare apps less when those apps rely on third party services to manage authentication or data processing, while 28.3% remained neutral. Only a small minority disagreed.

This suggests that third party involvement reduces user confidence, likely because participants feel less certain about who is handling their personal health information and how securely it is being processed.

What type of security do you usually use when logging into healthcare apps?
(Select all that apply)

59 responses

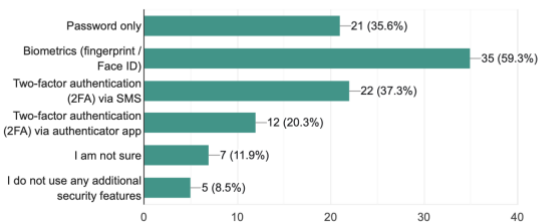


Figure 16: Security methods used when logging into healthcare applications

As shown in Figure 16, the most commonly used security method among participants was biometric authentication (59.3%), followed by two factor authentication (2FA) via SMS (37.3%), and password only login (35.6%).

A smaller proportion of participants reported using authenticator app based 2FA (20.3%), while 11.9% were unsure of the security methods they use and 8.5% reported not using any additional security features.

These findings suggest that while many users use stronger login methods such as biometrics and 2FA, a significant number still rely on weaker options or are unsure about their security settings. This highlights the need for better user awareness and stronger built in security features within healthcare applications.

8. Discussion

This section discusses the findings of this study and compares them with existing research to understand how users perceive the security and trustworthiness of healthcare applications.

The results show that healthcare applications are widely used, with most participants using them either regularly or occasionally. This reflects the growing reliance on digital healthcare systems identified in the literature [2]. However, despite this high usage, trust in these applications is not consistent. Only 53.3% of participants reported trusting healthcare apps to protect their personal health information, while 25% expressed outright distrust and 21.7% remained neutral. This contrasts with Belfrage et al. [16], who found that 81.9% of their respondents had high trust in healthcare systems to protect electronic patient data. This difference may be explained by the different contexts of the two studies. Belfrage et al. surveyed a general Swedish population about established healthcare systems, while this study focused on app-specific trust among a predominantly younger UK sample. It may also reflect growing public awareness of cyberattacks, as 52.6% of participants in this study reported that hearing about healthcare cyberattacks had reduced their trust in healthcare applications.

A significant finding was the gap between general awareness and specific understanding. While 86.6% of participants were aware that cyberattacks such as phishing and ransomware can affect healthcare services, only 71.7% understood what a healthcare data breach is, and only 41.7% felt they understood what information is stored in their digital health record. This distinction is important as it suggests that media coverage of cyberattacks may have raised general concern without improving practical knowledge. This aligns with the literature, which highlights that patient trust can be undermined not only when breaches occur, but also when individuals do not understand how their information is protected [15]. The cybok Human Factors framework similarly notes that limited security awareness increases vulnerability [24], which this study confirms from the user perspective.

Transparency and data sharing emerged as key factors affecting trust. Only 55% of participants knew who could access their medical records, and 56.7% reported trusting healthcare apps less when third-party services were involved in authentication or data processing. This supports findings in the literature, where users questioned whether data is “even stored safely... or passed on to third parties” [15] and is consistent with research showing that third-party providers are

frequently granted excessive access privileges rather than following the principle of least privilege [10]. The involvement of external services therefore appears to reduce both actual security and perceived security, compounding the trust problem.

The survey also revealed variation in authentication practices. While 59.3% of participants used biometric authentication and 37.3% used SMS-based two-factor authentication, 35.6% still relied on password-only login and 8.5% reported using no additional security features at all. This is concerning given the NCSC's guidance that MFA provides significantly stronger protection than passwords alone [25]. The continued reliance on weaker authentication among a substantial minority of users highlights the gap between recommended security standards and real-world behaviour a gap that the cybok Human Factors knowledge area attributes to security fatigue and the prioritisation of convenience over security [24].

There was also a slight but inconclusive preference for NHS applications over private ones, with 45% of participants trusting NHS apps more compared to 23% who disagreed. However, the high neutral response of 31.7% suggests that many users do not distinguish clearly between public and private healthcare platforms when it comes to trust. This may reflect limited understanding of how different platforms manage data rather than a genuine evaluation of their security models, which is consistent with the wider finding that transparency is insufficient across both public and private healthcare applications.

Overall, the findings suggest that while healthcare applications are increasingly relied upon, there are significant gaps in user understanding, trust, and security behaviour that leave both patients and systems vulnerable. These gaps align with the vulnerabilities identified in the literature weak authentication, insufficient transparency, third-party risk, and limited security awareness suggesting that the current state of healthcare cybersecurity resilience, from the user perspective, remains insufficient.

This study has limitations that should be acknowledged. The sample was predominantly younger participants aged 18 to 24, comprising 78.3% of respondents, which limits the generalisability of the findings to older age groups who may have different levels of digital literacy and different experiences of healthcare applications. The sample size of 61 participants is also relatively small for drawing broad conclusions. Future research with a more diverse and larger sample would strengthen the reliability of these findings.

9. Conclusion

Summary

This project aimed to examine how resilient healthcare systems are in protecting patient data, and to explore how much the public trusts and understands cybersecurity in healthcare applications. Based on the findings, the healthcare system is not yet fully resilient, despite the protections that exist.

The literature review showed that healthcare systems face serious and ongoing cyber threats, including ransomware, phishing, and data breaches. The wannacry attack is a clear example of how even when a security fix is available, it does not always get applied in time, leaving systems vulnerable. Human behaviour also plays a role, as staff and users often bypass security measures due to time pressure or lack of awareness.

The survey findings supported this. Most participants used healthcare apps regularly, but trust was inconsistent only 53.3% trusted apps to protect their data, while many others were unsure or did not trust them at all. Many participants also did not fully understand how their data is stored or who can access it, and a significant number still used weak login methods such as passwords only. This shows that the problem is not just technical but also about awareness and communication.

Overall, while progress has been made, healthcare systems still have a long way to go before they can be considered truly resilient. Improving transparency, encouraging stronger authentication, and raising awareness among both users and staff are key steps needed to better protect patient data and rebuild trust.

Recommendations for further work

Several improvements could be made to strengthen this project. Firstly, although a range of age groups was included, the sample was mostly made up of younger participants. Future research should aim to include a more balanced range of ages to improve the generalisability of the findings.

In addition, the sample size was relatively small, which may limit how representative the results are. Increasing the number of participants would make the findings more reliable.

Another improvement would be how the survey was distributed. It was shared through limited channels, which may have reduced the diversity of participants. In future, it could be shared across a wider range of platforms to reach a broader audience.

Future work could also focus on improving user awareness of healthcare cybersecurity through clearer privacy policies, better communication, and education on data protection.

Finally, further research could explore how security features such as biometric authentication or multi-factor authentication impact user trust.

Reflection

For this project, I chose to use a survey because cybersecurity is a broad and often technical field, but I wanted to take a simpler and more practical approach. I focused on everyday users and how cybersecurity affects people in their daily lives, especially those who are not very technical. This helped me understand real user opinions rather than just focusing on technical concepts.

One thing I learned is that people's responses are not always clear or confident. Many participants gave neutral answers, which showed a lack of understanding rather than strong opinions. This made me realise the importance of capturing natural, non-robotic responses, as people's views are influenced by their experiences and emotions, not just technical knowledge.

When distributing the survey, I mainly used platforms like linkedin, but I realised that using social media platforms such as Instagram or Snapchat could have helped me reach a wider and more diverse audience.

Looking back, I think combining surveys with interviews would have improved the project. Interviews could have provided deeper insights into how people feel and think, rather than relying only on fixed responses. Overall, this project helped me understand that cybersecurity is not just technical, but also about user trust and perception.

Overall, I believe I worked well throughout the project and stayed consistent with my aims. I was able to meet my main goals of collecting data and understanding user trust in healthcare applications. However, I did leave some parts of the project later than I should have, which affected how much time I had to improve areas such as survey distribution and sample diversity. This meant I was not fully able to achieve a wide and representative sample. If I had managed my time

better, I could have planned the data collection stage more effectively and reached a broader audience.

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11.Appendices

Ethical form

68 4/13/2026 3:03 4/13/2026 3:13 mouness@roehampton.ac.uk
 Sawsan Mounes Sawsan Mounes MOU23620003 Cyber Security
 Undergraduate Final Year Project (CMP-X301-0) 3 4/13/2026 How

Resilient Is the HealthCare System In Protecting Patient Data? "I chose this topic because I'm interested in the healthcare industry and I've seen a lot of news about cyberattacks and data breaches affecting it. That made me realise how vulnerable it can be, which was a bit of a lightbulb moment for me.

I also prefer more interaction with people, similar to customer service roles, rather than purely technical work behind the scenes. Because of this, I wanted to focus on how real users think and feel about healthcare apps and security.

To explore this, I created a survey to collect people's opinions on data security, privacy, and cyber threats. The main output of this project is the survey and the analysis of the results, which highlights gaps in user awareness and trust." Yes Survey

(online) emails , sharing ,asking around in class , linkedin ,social media , family

Both If Roehampton students or staff participate in my research, I will ensure that I have explained the project carefully, obtained their verbal consent and explained that they may withdraw from the study at any time.;If my research involves people outside Roehampton University I will ensure that I have obtained written consent for their participation and confirmation of their right to withdraw.;All questionnaires or surveys will include participant information and a consent option;The data collected will only be used for the purpose of this study;I will anonymise all data collected from individuals in my report;I will hold data obtained for research only on Roehampton University's OneDrive ;If I make any video or audio recordings, these will be held securely, not used for any purpose other than this research;All data collected, including audio and video recordings, will be deleted when I cease to be a Roehampton University student;If any activities may require a risk assessment I will speak with my Supervisor and prepare this.;I will not be carrying out any research directly with under 18 year olds or vulnerable adults;I will not be recording any 'special category' information about individuals;I can confirm that all Information supplied on this form is correct and my work complies with ethical standards

Supervisor: Dr Mastaneh Davis



Associate Professor in Cyber Security,

Programme Leader BSc(Hons) Cyber Security
Head of Learning and Teaching for Computing
School of Computing, Engineering and the Built Environment

Project Proposal poster



Risk register

Risk	Description	Impact	Likelihood	Mitigation
Low survey responses	Not enough participants complete the survey	High	Medium	Keep survey short, simple, and easy to complete
Misunderstanding questions	Participants may not understand cybersecurity terms	Medium	Medium	Use simple language and test questions before sharing
Limited participant range	Mostly younger participants respond	High	High	Share survey on multiple platforms to reach different age groups
Data privacy concerns	Participants may worry about how their data is used	High	Low	Keep survey anonymous and clearly explain consent
Time management issues	Leaving tasks too late may affect quality	Medium	Medium	Plan tasks earlier and set deadlines
Biased responses	Participants may not answer honestly or may rush	Medium	Low	Keep survey clear and not too long to encourage genuine answers
Technical issues	Problems accessing or submitting the survey	Low	Low	Test survey before distribution

Appendix A

Informed Consent Form

Title : How Resilient Is the HealthCare System In Protecting Patient Data?

Researcher : Sawsan Mounes

Institution : University of Roehampton

Part 1 : Information Sheet

You are invited to take part in a research study about public awareness, trust, and confidence in the security of healthcare applications (such as the NHS App or Patient Access).

The purpose of this study is to understand how people perceive data security, transparency, and trust in healthcare apps.

Your participation involves completing an online questionnaire with approximately 12–15 questions. The survey will take around 5–10 minutes to complete.

Participation is voluntary. You do not have to take part, and you may stop at any time without giving a reason.

No personal, medical, or identifying information will be collected. All responses are anonymous and will be used only for academic purposes.

The data collected will be stored securely and handled in line with GDPR and HIPAA principles.

If you have any questions about this study, you may contact the researcher before completing the survey.

Data Collection and Confidentiality

- No personal, medical, or identifying information will be collected.
- All responses are anonymous.
- The data collected will be used only for academic purposes as part of this university project.

Data Storage and Retention

- Survey data will be stored securely on password-protected systems accessible only to the researcher.
- The data will be retained only for the duration of the project and any required academic assessment.

Data Deletion and Disposal

- Once the project and assessment process are complete, all collected data will be permanently deleted.
- Any temporary files or backups created during analysis will also be removed.
- If removable storage were to be used, it would be fully reformatted to ensure that data cannot be recovered.

Risks and Benefits

There are no anticipated risks associated with participation.

There is no direct benefit to participants; however, the study may contribute to a better understanding of trust and security in healthcare applications.

Part 2 : Consent Statement

By continuing with this survey, you confirm that:

- You are 18 years or older
- You have read and understood the information above
- You voluntarily agree to take part in this study
- You understand that your responses are anonymous

☐ I consent to participate in this research study

I have read the foregoing information, or it has been read to me. I have had the opportunity to ask questions about it and any questions I have been asked have been answered to my satisfaction. I consent voluntarily to be a participant in this study

Print Name of Participant _____

Signature of Participant _____

Date _____

Email(to send survey) _____

Statement by the researcher/person taking consent

I confirm that the participant was given an opportunity to ask questions about the study, and all the questions asked by the participant have been answered correctly and to the best of my ability. I confirm that the individual has not been coerced into giving consent, and the consent has been given freely and voluntarily.

A copy of this ICF has been provided to the participant.

Print Name of Researcher/person taking the consent _____

Signature of Researcher /person taking the consent _____

Date _____

Appendix B

Part 1 : Participant Information and Consent

You are invited to take part in a research study about public awareness, trust, and confidence in the security of healthcare applications (such as the NHS App or Patient Access).

Your participation involves completing an online questionnaire with approximately 12–15 questions. The survey will take around 5–10 minutes.

Participation is voluntary. You may stop at any time.

No personal, medical, or identifying information will be collected. All responses are anonymous and used for academic purposes only.

Survey data will be stored securely and deleted after the project is completed. There are no anticipated risks associated with participation.

Data Collection and Confidentiality

- No personal, medical, or identifying information will be collected.
- All responses are anonymous.
- The data collected will be used only for academic purposes as part of this university project.

Data Storage and Retention

Survey data will be stored securely on password-protected systems accessible only to the researcher.

The data will be retained only for the duration of the project and any required academic assessment.

Data Deletion and Disposal

Once the project and assessment process are complete, all collected data will be permanently deleted.

Any temporary files or backups created during analysis will also be removed.

If removable storage were to be used, it would be fully reformatted to ensure that data cannot be recovered.

Risks and Benefits

There are no anticipated risks associated with participation.

There is no direct benefit to participants; however, the study may contribute to a better understanding of trust and security in healthcare applications.

Participants will not receive individual copies of responses, as the survey is conducted anonymously

1. **Are you 18 years of age or older? ***

Mark only one oval.

☐ Yes

☐ No

2. **Please confirm the following statements: ***

Check all that apply.

☐ I have read and understood the participant information provided above.

☐ I understand that participation is voluntary.

☐ I understand that my responses are anonymous.

☐ I understand that no personal or medical information is being collected.

3. **Do you consent to participate in this research study? ***

Mark only one oval.

☐ Yes, I consent

☐ No, I do not consent

Kanban Board

